Composite rebar for durable and sustainable concrete infrastructure

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Background

The long-term performance of reinforced concrete structures is frequently governed by the durability of the embedded reinforcement rather than by the compressive strength of concrete itself. In aggressive environments, corrosion of steel reinforcement induces cracking, delamination and spalling of the concrete cover. This deterioration mechanism compromises structural integrity, increases maintenance requirements and shortens service life. Growing demands for extended design life, reduced lifecycle cost and lower environmental impact have accelerated the development of non-metallic reinforcements. Amongst these, fibre-reinforced polymer (FRP) composite rebar has emerged as an alternative to conventional steel reinforcement specifically in applications where corrosion resistance is critical.

Materials and manufacturing process

Composite rebar is a non-metallic reinforcement system consisting of continuous high-strength fibres embedded within a thermosetting polymer matrix. In structural applications, glass fibres (GFRP) are the most widely used reinforcement due to their balanced mechanical performance, corrosion resistance and cost efficiency. Basalt fibres (BFRP) serve as an alternative where improved chemical stability and higher temperature resistance are desired, while carbon fibres (CFRP) are selected in applications requiring higher stiffness, superior fatigue resistance or reduced creep under sustained loading. Within this composite architecture, the fibres act as the primary load-carrying phase, providing tensile strength and stiffness, while the polymer matrix binds the fibres together, facilitates stress transfer between individual filaments and protects the reinforcement from moisture, chemical exposure and other environmental effects.

The performance of composite rebar is strongly influenced by the selection of the polymer matrix system. Thermosetting resins commonly employed in structural FRP reinforcement include epoxy, vinyl ester and unsaturated polyester resins. These matrix systems differ in mechanical properties, chemical resistance, curing behaviour and long-term durability under environmental exposure. The matrix plays a critical role in ensuring adequate fibre wet-out during impregnation, maintaining fibre alignment and dimensional stability during curing and enabling effective stress transfer at the fibre-matrix interface. Its resistance to moisture ingress, alkaline environments and chemical attack is essential for preserving the mechanical integrity of the composite over the intended service life.

Amongst thermosetting matrix systems, epoxy resins are frequently

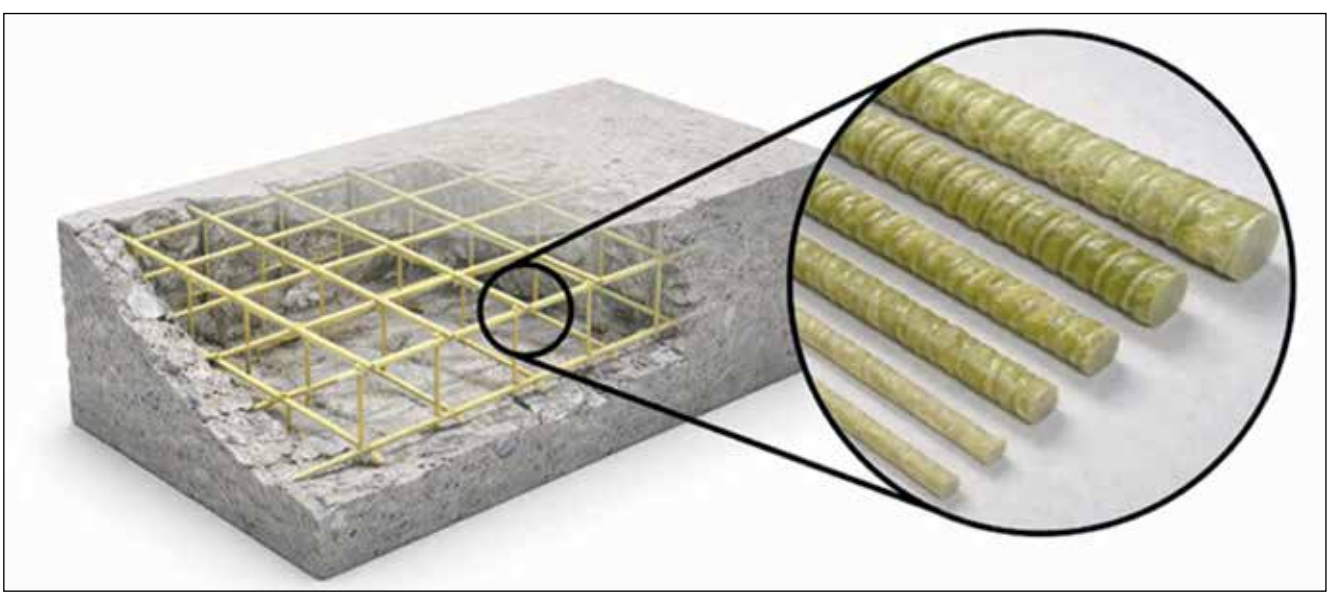

Fig. 1: Fibre-reinforced polymer (FRP) composite rebar embedded in concrete

preferred for high-performance composite rebar due to their strong interfacial adhesion, low curing shrinkage and stable crosslinked network structure. Compared with polyester-based systems, epoxies typically provide higher mechanical strength and improved durability under hydrothermal and chemically aggressive conditions, while promoting efficient stress transfer at the fibre-matrix interface. In addition, epoxy formulations are free of volatile organic compounds (VOCs), particularly compared with styrene-containing systems, resulting in increased safety and reduced emissions during processing. Their tuneable curing behaviour further enables precise process control in continuous manufacturing and supports consistent long-term performance.

To accommodate different manufacturing strategies and processing conditions, *Epotec*® epoxy systems from Aditya Birla Advanced Materials, for composite rebar production are available as multi-component formulations. These include:

- 2K components: resin base and curing agent.
- 3K components: resin base, curing agent and accelerator.
- 4K components: resin base, curing agent, accelerator and additional functional modifiers (e.g., plasticizers or flexibilizers).

The separation of functional components enables independent adjustment of pot life, viscosity, curing kinetics and crosslink density, thereby allowing precise control over impregnation behaviour and final mechanical properties. Such formulation flexibility is particularly important in continuous manufacturing process, where stable fibre wet-out, controlled gelation and consistent curing are required to achieve uniform fibre volume fraction and reliable mechanical performance.

In industrial production, unidirectional longitudinal fibre bundles are first impregnated with a thermosetting resin to ensure uniform wet-out and controlled fibre volume fraction. The impregnated core is subsequently combined with additional fibre bundles that are helically wrapped around the longitudinal reinforcement prior to entering the heated forming zone. The entire assembly is then continuously drawn through a temperature-controlled curing chamber, where thermal crosslinking consolidates the structure into a rigid, integrated composite profile. Post-curing surface treatments, such as sand coating or geometric profiling, may be applied to further enhance bond interaction with concrete. Process parameters including fibre alignment, wrapping tension, resin viscosity, temperature profile and curing kinetics directly govern mechanical performance, bond behaviour and long-term durability.

Comparison of steel and composite rebar

Steel and composite rebars exhibit fundamentally different mechanical behaviour that directly influences their structural performance in reinforced concrete. Conventional steel rebar is characterised by a distinct yield point followed by plastic deformation, providing ductility and warning prior to failure, with typical yield strengths of 400-600 MPa and an elastic modulus of approximately 200 GPa (ASTM A615; Nilson *et al*., 2010). In contrast, composite rebars, including GFRP, BFRP and CFRP, exhibit higher ultimate tensile strength, typically ranging from 600 to over 1,500 MPa, but lower elastic modulus and linear-elastic response up to brittle failure (ACI 400.1R). Consequently, serviceability criteria often govern design, as prescribed in FRP

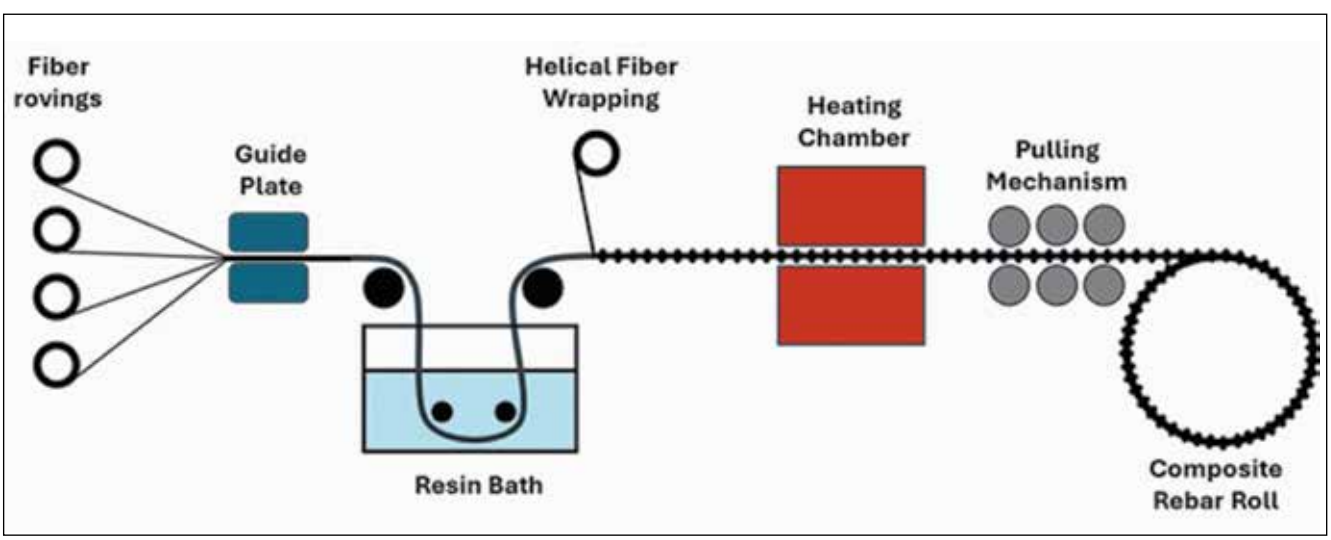

Fig. 2: Schematic illustration of the manufacturing process for composite rebar

design recommendations, while material property requirements are specified in ASTM D7957.

In addition to their mechanical characteristics, FRP composite rebars offer significant durability and functional advantages. They are inherently resistant to electrochemical corrosion and do not rust, thereby reducing the risk of corrosion-induced cracking and spalling in chloride-laden or chemically aggressive environments. Furthermore, GFRP and BFRP rebars are electrically non-conductive and non-magnetic, mitigating stray currents, galvanic effects and electromagnetic interference in electrically sensitive infrastructure. CFRP rebars, depending on fibre architecture, may exhibit partial electrical conductivity and are therefore less suitable where electrical insulation is required.

Within the context of decarbonization strategies and net-zero infrastructure development, composite rebar is increasingly evaluated using life cycle assessment (LCA) to quantify its environmental performance relative to conventional steel reinforcement. Although FRP reinforcement may exhibit higher embodied energy and carbon emissions at the production stage compared with steel, reported LCA results indicate that its whole-life greenhouse gas emissions can be lower when durability and service life are incorporated into the assessment boundary. Cradle-to-grave analyses have shown that GFRP rebars can achieve approximately 17% lower CO_2-equivalent emissions per kilogram over their life cycle compared with conventional steel reinforcement. When evaluated at the structural or project level, accounting for reduced material demand and lower transportation impacts due to lower density, total lifecycle CO_2-equivalent emissions have been reported to decrease by approximately 78% to 85% relative to comparable steel-reinforced configurations (Al Omar and Abdelhadi, 2024).

Reduced mass also lowers transportation energy and handling impacts and the corrosion resistance of FRP rebars significantly reduces maintenance, repair and premature replacement over the service life of concrete structures, further contributing to improved lifecycle environmental performance.

Applications of composite rebar

Composite rebar is primarily applied in infrastructure where corrosion resistance, durability and functional performance are critical design drivers. Typical applications include bridge decks exposed to de-icing salts, marine and coastal structures, offshore concrete elements in chloride-rich environments, wastewater treatment facilities and industrial floors subjected to aggressive chemicals. In addition, its non-conductive and non-magnetic properties make it particularly suitable for electrically sensitive infrastructure such as railway systems, tunnels, substations, data centres and medical imaging facilities where mitigation of stray currents and electromagnetic interference is required. The low density of composite reinforcement further supports its use in precast elements, modular construction and remote installations by reducing transportation and handling demands. Through the combination of corrosion resistance, electrical neutrality and weight reduction, composite rebar extends service life and reduces maintenance requirements in demanding service environments.

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